

Huawei Certified ICT Associate-5G

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Module Questions

5G Motivation and Industry Progress

1. (True or False) The formal name of 5G is IMT-2020, which was defined by ITU-R.
 - **True**
 - False
2. (Single choice) For eMBB scenario of 5G, the peak rate is required to reach ().
 - 1 Gbps
 - **10 Gbps**
 - 100 Gbps
 - 1000 Gbps
3. (Single choice) Based on the vision defined by ITU on 5G, how many connections should be supported in one square kilometer for mMTC scenario?
 - 1000
 - 10,000
 - 100,000
 - **1,000,000**
4. (Multiple choices) Based on the developing trend, what new characteristics could 5G network have?
 - **Higher rate**
 - **Stronger network connectivity capability**
 - **Lower latency**
 - More complex network structure
5. (Multiple choices) To realize the general vision on 5G, what application scenarios have been defined by ITU?
 - **eMBB**
 - **mMTC**
 - VoIP
 - **uRLLC**

5G Network Architecture and Key

1. (True or False) uRLLC cannot be supported by NSA mode.
 - **True**
 - False
2. (True or False) Sub6G Hz includes sub3G Hz and C-band.
 - **True**
 - False

3. (Single choice) In current eMBB scenario, what channel coding method is adopted on control channels of 5G air interface?
 - LDPC
 - Turbo
 - Convolutional code
 - **Polar**
4. (Multiple choices) What gains can be realized by massive MIMO of 5G?
 - **Beam forming**
 - **UL MU-MIMO**
 - **DL MU-MIMO**
 - **SU-MIMO**
5. (Multiple choices) Which ones of the below techniques can be used to solve the uplink and downlink imbalance problem of 5G radio network?
 - **Uplink and downlink decoupling (SUL)**
 - **Dual connection**
 - CU and DU separation
 - LDPC coding

5G Use Case

1. (True or False) Considering the safety of automated driving service, it is required to control the latency of mobile network on nano-second level.
 - True
 - **False**
2. (True or False) The vertical industries have clear requirement on service isolation. This requirement can be satisfied by network slicing technology.
 - **True**
 - False
3. (True or False) For intelligent grid service, power transmission and power generation will be the main application field for 5G.
 - True
 - **False**
4. (Multiple choices) Based on the ultra low latency of 5G network, what new services can 5G support?
 - **Remote health care**
 - Automated driving
 - **Intelligent manufacturing**
 - HD voice
5. (multiple choices) Based on 5G network, what benefits can CloudVR bring comparing with the traditional PC VR?
 - **Effective control on copyright**
 - **Lowering the VR terminal cost**
 - **Better experience on mobility**
 - Increasing the VR terminal prices

Typical Industry Application Solutions of

1. (True or False) C-V2X is a global standard communication technology specified by 3GPP, which includes LTE-V2X and 5G-V2X
 - **True**
 - False
2. (True or False) DSRC provides short-distance wireless transmission technology based on IEEE802.16, which is mainly used to vehicle-to-vehicle and vehicle-to-road communication.
 - True
 - **False**
3. (Single choice) Which of the below key technologies of 5G can effectively support the precise load control of intelligent grid?
 - Massive MIMO
 - Base station cloudification
 - **Edge computing**
 - F-OFDM
4. (Multiple choices) The typical valued application scenarios of remote driving in 5G includes ().
 - **Campus shuttle bus**
 - **Port transportation**
 - **Mining area transportation in harsh environment**
 - **Vehicle resource scheduling**
5. (Multiple choices) Why is 5G technology required for automated driving?
 - **5G can provide lower latency**
 - **It is not restricted by the distance and environment**
 - **Promoting the evolution from single-vehicle intelligence to collaborative intelligence**
 - **Better interactive experience**

Final Exam

True or False

30 points possible (graded)

1. To enhance the spectrum efficiency, massive MIMO technology is introduced in 5G network, whose maximum number of antennas may reach 64T64R.
 - **True**
 - False
2. C-band is the main frequency band of 5G. Currently, a maximum of 100 MHz bandwidth can be deployed. Uplink and downlink imbalance can be resolved by the uplink and downlink decoupling feature.
 - **True**
 - False

3. The 5G network features high rate, low latency, and massive connections, and can better support vertical industry applications. - **True** - False

4. Based on the massive MIMO beam forming capability, 5G base stations can better support low-altitude coverage.

- **True**
- False

5. The SA mode of 5G network favors aggressive operators on fast deployment of 5G services based on 4G network.

- True
- **False**

6. For intelligent grid service, power transmission and power generation will be the main application field for 5G.

- True
- **False**

7. The IEEE is responsible for 5G standardization, and formulates and releases 5G mobile communication technical specifications and reports.

- True
- **False**

8. Because of the high frequency band of millimeter wave, the penetration loss of it is very small.

- True
- **False**

9. mMTC can be supported by both NSA and SA mode.

- True
- **False**

10. In IoV scenarios, the DSRC technology provides better coverage than C-V2X.

- True
- **False**

Multiple Choices

11. Which of the following are CloudRAN benefits?

- **Unified architecture, achieving multi-dimensional convergence of multiple RATs, frequency bands, multi-layer networks, and ultra-dense networks.**
- **On-demand deployment, intelligent slicing, and adaptation to diverse services, such as large bandwidth, short latency, and massive connections.**
- **Open platform and intelligent O&M, promoting agile service rollout.**
- **Resource pooling, improving resource utilization and elastic network capacity expansion.**

12. In the future, the 5G-V2X technology can be used to enable high-level autonomous driving applications. Which of the following are included?

- **Vehicle platooning**
 - *Advanced driving**
 - **Sensor information sharing**
 - **Remote driving**
13. Based on the C-V2X technology, multi-party collaboration among transportation participants can be implemented to facilitate intelligent transportation. The C-V2X includes:
- **V2V**
 - **V2P**
 - **V2N**
 - **V2I**
14. Compared with 4G networks, which ones of the following services can be better supported by 5G networks that support higher rates?
- **4K and 8K HD video**
 - **VR**
 - **AR**
 - **Holographic image service**
15. For 5G network, what are the benefits of using a globally unified standard?
- **Supporting global roaming**
 - **Economies of scale**
 - **Reducing industry chain risks**
 - Increasing complexity of mobile phones
16. With the rapid development of mobile services and the emergence of new services, what are the challenges faced by mobile communications networks?
- **Rapid growth of mobile data volume**
 - **The number of network connections keeps increasing**
 - **Diversified service scenarios**
 - Rapid development of video call service
17. Which ones of the following statements about millimeter waves are incorrect?
- **The lower the frequency band is, the better the coverage is.**
 - **Newly introduced in NR which has the maximum cell bandwidth.**
 - **The available frequency resource is limited, most of which was occupied by current systems.**
 - **Coverage capability is poor and requirements on RF components are high.**
18. Which ones of the below 5G core network functions belong to control plane?
- **AMF**
 - UPF
 - **SMF**
 - **PCF**

Single Choice

19. Which of the following statements about the 5G protocol standardization process is correct?
- **The 5G protocol is divided into two phases: Phase 1 and Phase 2, which correspond to two protocol versions: R15 and R16.**

- 3GPP Release 16 focuses on the standardization of eMBB service scenarios.
- 3GPP Release 15 focuses on the standardization of mMTC service scenarios.
- Based on the current protocol progress, 5G will be put into commercial use globally in 2025.

20. Which one of the following scenarios are the first batch of 5G network applications mainly focused on?

- **eMBB**
- mMTC
- uRLLC
- ALL

21. Which of the below frequency bands is the C-Band supported by 5G?

- 2600MHz
- **3.5GHz**
- 28G
- 700MHz

22. For eMBB scenario of 5G, the peak rate is required to reach ().

- 1 Gbps
- **10 Gbps**
- 100 Gbps
- 1000 Gbps

23. Which of the following functions can be tailored for smart grid meter reading services?

- Charging function
- **Mobility management**
- Session management
- Terminal management

24. Which of the below statements about 5G NSA networking is incorrect?

- The signaling of the UE can be forwarded only by the eNodeB.
- **The UE signaling can be directly sent to the gNodeB.**
- ##Downlink user plane data can be sent to UE through gNodeB.
- Downlink user plane data can be sent to UE through eNodeB.

25. Which RAN structure will be separated into CU and DU two parts?

- C-RAN
- D-RAN
- **Cloud RAN**
- All of them

26. Which of the below key technologies of 5G can effectively support the precise load control of intelligent grid?

- Massive MIMO
- Base station cloudification
- **Edge computing**
- F-OFDM

27. Which of the following technologies can shorten the air interface latency?
- High-level Modulation
 - Massive MIMO
 - **Flexible frame structure**
 - F-OFDM
28. 5G network can use high-order modulation to increase the air interface rate. Which of the following statements about high-order modulation is incorrect?
- Higher-order modulation can improve spectrum utilization.
 - Higher-order modulation has higher requirements on the SINR of signals.
 - High-order modulation technologies increase terminal costs.
 - **256QAM and massive MIMO cannot be used together.**
29. Which of the following statements about the causes of insufficient 5G uplink coverage is incorrect?
- The UE transmit power is low.
 - No beamforming is used in the uplink.
 - Uplink slot assignment is few.
 - **UE receiver sensitivity is poor.**
30. In 5G system, spectrum utilization can be improved by using the F-OFDM technology. When the SCS is 30 kHz, which system bandwidth is of the maximum spectrum utilization?
- 20MHz
 - 60MHz
 - 80MHz
 - **100MHz**